

Key

Review for Test #1 (Chapter 1)

In Exercises 1–4, use the diagram.

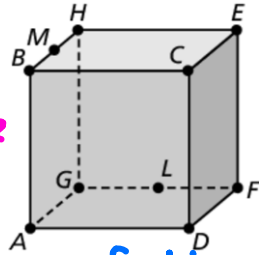
L or F
EF
m

1. Name a point that is coplanar with points A, D, and G.

2. Name the intersection of plane HEG and plane DFE. ** ← MUST be a line **

3. Name a point that is collinear with $\overline{BH}$.

4. Name a point that is not coplanar with points C, E, and M. *→ any of the following A, G, L, F, or D*



In Exercises 5 and 6, the endpoints of $\overline{JK}$ are given.

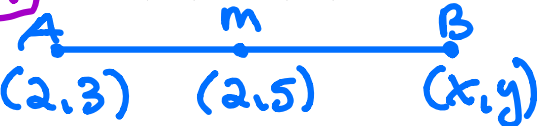
Find the coordinates of the midpoint M.

5. $J(-3, 2)$ and $K(9, 2)$ *$\frac{(-3+9)}{2}, \frac{(2+2)}{2}$*

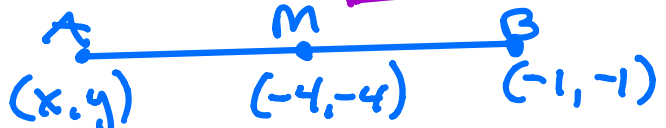
6. $J(1, 3)$ and $K(7, 5)$ *$\frac{(1+7)}{2}, \frac{(3+5)}{2}$*

In Exercises 7 and 8, the midpoint M and one endpoint of $\overline{AB}$ are given. Find the coordinates of the other endpoint.

7. $M(2, 5)$ and $A(2, 3)$



8. $M(-4, -4)$ and $B(-1, -1)$



In Exercises 9 and 10, find the distance between the two points.

9. $Q(5, 6)$ and $P(1, 3)$

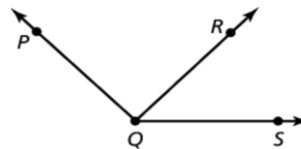
$$\sqrt{(5-1)^2 + (6-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

10. $G(2, 5)$ and $H(4, -1)$

$$\sqrt{(2-4)^2 + (5-(-1))^2} = \sqrt{4 + 36} = \sqrt{40} \approx 6.3$$

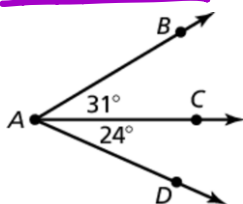
11. Name three different angles in the diagram.

$\angle PQR, \angle RQS, \angle PQS$



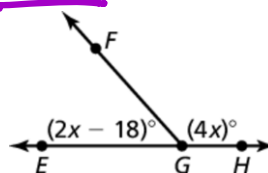
In Exercises 12 and 13, find the indicated angle measure.

12. Find $m\angle BAD$.



$$31 + 24 = 55$$

13. Find x.



$$x = 33$$

$$2x - 18 + 4x = 180$$

$$6x - 18 = 180$$

$$6x = 198$$

In Exercises 14 and 15, find the measure of each angle.

14. $\angle WXY$ and $\angle YXZ$ are supplementary angles, $m\angle WXY = (6x + 59)^\circ$, and $m\angle YXZ = (3x - 14)^\circ$.

$$m\angle WXY = 149^\circ$$

$$m\angle YXZ = 31^\circ$$

$$6x + 59 + 3x - 14 = 180$$

$$9x + 45 = 180$$

$$9x = 135$$

$$x = 15$$

15. $\angle ABC$ and $\angle CBD$ are complementary angles, $m\angle ABC = (3x + 6)^\circ$, and $m\angle CBD = (4x - 14)^\circ$.

$$3x + 6 + 4x - 14 = 90$$

$$7x - 8 = 90$$

$$x = 14$$

$$7x = 98$$

$$m\angle ABC = 48^\circ$$

$$m\angle CBD = 42^\circ$$

16. Two angles form a linear pair. The measure of one angle is 24° more than the measure of the other angle.

$$78^\circ, 102^\circ$$

$$(x + 24) + (x) = 180$$

$$x = 78$$

$$2x + 24 = 180$$

17. The measure of an angle is three times the measurement of its complement.

$$67.5^\circ, 22.5^\circ$$

$$x = 3(90 - x)$$

$$x = 270 - 3x$$

$$4x = 270$$

OR

$$x = 67.5$$

$$x + 3x = 90$$

$$4x = 90$$

$$x = 22.5$$

18. The measure of one angle is 15 less than half the measurement of its supplement.

$$50^\circ, 130^\circ$$

$$x = \frac{1}{2}(180 - x) - 15$$

$$x = 90 - \frac{1}{2}x - 15$$

$$\frac{3x}{2} = 75$$

$$x = 50$$

OR

$$\left(\frac{1}{2}x - 15\right) + (x) = 180$$

$$\frac{3}{2}x - 15 = 180$$

$$\frac{3}{2}x = 195$$

$$x = 30$$